A Coupled Quadratic Reserve for Spectral Scalar Curvature

Physical Graph Residuals and the Conditional Uniform-Width Interface

Dr. Muralidhara Subbarao
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www.integralresearch.net
Stony Brook University
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Abstract

The assigned manuscript converts a lower bound for the physical quadratic form of $-4\Delta+\mathrm{Scal}$ into a signed-drift curvature reserve, then uses local metric enlargements and a separate partition theorem to obtain a uniform codimension-two width bound. We retain that distinction. For its actual ordinary-graph preprocessing equation, changing the linear discrepancy pressure from 10 to 2 and completing the pressure square together with the output drift gives the reserve cost $4n + 4/3$, in place of $100n + 203$. The proof needs no uniform graph-angle bound. The same calculation yields a whole-domain residual certificate with added cost $2\varepsilon/3 + \varepsilon^2/3 + 2\varepsilon_1$ when the equation defect has absolute value at most $\varepsilon \le 1$ and gradient norm at most ε_1. A fixed-angle oscillatory graph shows why the derivative condition cannot simply be dropped. We explain the genuine noncompact existence argument, the signed pressure law, the locally finite assembly and the physical form identity, without promoting finite tests to a global width proof. Contribution assessment: C3/5, a source-specific coupled reserve improvement and slope-independent residual extension, with the global stretching and partition machinery explicitly retained as inputs.

Contents

1 Problem, normalization, and the source boundary **2**
1.1 Architecture actually read 2

2 The actual graph equation, not a finite inverse **3**

3 A coupled square completion **4**

4 An off-equation certificate without a global angle bound **5**

5 Actual graph examples and a necessary derivative budget **7**

6 Physical forms and what a finite Ram certificate can establish **7**

7 Signed pressure and the unchanged global interface **8**
7.1 Why the sign of the drift matters 8
7.2 Trace losses, local finiteness, and initialization 9
7.3 What still gives width 10

8 Status, advantages, and remaining work **10**

9 Read scope and reproducible checks **10**

1 Problem, normalization, and the source boundary

Let (M^n, h) be a connected smooth Riemannian manifold without boundary. Our graph calculation holds for $n \geq 1$; the assigned width problem assumes $n \geq 4$. Set $\Delta_h = \operatorname{div}_h \nabla_h$, and for real smooth t write

$$\mathcal{D}(h,t) = \operatorname{Scal}_h + 2\Delta_h t - |\mathrm{d}t|_h^2, \qquad \operatorname{div}_{h,t} X = \operatorname{div}_h X - \mathrm{d}t(X). \tag{1}$$

Neither t nor its gradient is assumed globally bounded. All unqualified integrals use the stated Riemannian volume; weighted integrals explicitly display e^{-t}. A complete metric is essential in the source's global exhaustion and width argument, not in the pointwise algebra.

The manuscript [1], dated October 5, 2026, asserts that for each $n \geq 4$ there is a dimensional constant C_n such that every complete (M^n, g) satisfying

$$\int_M (4|\mathrm{d}\phi|_g^2 + \operatorname{Scal}_g \phi^2)\,\mathrm{d}\operatorname{vol}_g \geq \int_M \phi^2\,\mathrm{d}\operatorname{vol}_g \qquad (\phi \in C_c^\infty(M;\mathbb{R})) \tag{2}$$

admits a continuous map to a simplicial complex of dimension at most $n-2$, every entire fiber of which has g-diameter at most C_n. This is not merely a bound on the connected components of fibers. No orientability, spin, compactness, or bounded-geometry hypothesis is added here. A spectral lower bound $\lambda > 0$ instead of 1 is normalized by replacing g with λg; distances scale by $\sqrt{\lambda}$.

On a noncompact manifold (2) does not assert that the spectral bottom is an attained L^2 eigenvalue. The source obtains a positive smooth supersolution v by normalized Dirichlet solutions on an exhaustion, local Harnack and Schauder bounds, and diagonal extraction. With $w = -2\log v$,

$$\mathcal{D}(g,w) = \frac{(-4\Delta_g + \operatorname{Scal}_g)v}{v} \geq 1. \tag{3}$$

These are classical ground-state principles, not new Ram transform identities. The construction supplies no global bound on w.

1.1 Architecture actually read

All seven public mathematical TeX files, the bibliography and README were read in full, with the intervals and hashes recorded separately. Their roles are as follows.

Module	Mathematical role and retained boundary
Introduction/main	Physical spectral hypothesis, noncompact theorem and scale.
Preprocessing	Positive supersolution; weighted graph angle equation; global graph; quadratic reserve.
Pressure	Inverse construction for a discrepancy pressure valid for every signed drift.
Stretching	Local graph with confined height, metric enlargement, scalar comparison and cutoff transition.
Assembly	Trace-decrease loss budget, locally finite changes, rescaling rounds and countable log warps.
Interface	Imported partition theorem, separation targets and whole-fiber diameter conclusion.

Full literal reading does not constitute an independent certification of all those global arguments. The new results below concern the ordinary preprocessing graph and its residual. The later pressure, stretching and partition conclusions are dependencies, not consequences of our tests.

2 The actual graph equation, not a finite inverse

For $w \in C^\infty(M)$ and a fixed $c > 0$, consider

$$\operatorname{div}_{h,w}\left(\frac{\nabla_h u}{\sqrt{1+|\mathrm{d}u|_h^2}}\right) = c(u-w). \tag{4}$$

For any smooth candidate u put

$$s = |\mathrm{d}u|_h, \quad l = (1+s^2)^{-1/2}, \quad \beta = \log l, \quad h^+ = h + \mathrm{d}u \otimes \mathrm{d}u, \quad t_u = w - \beta, \quad t^+ = t_u - u. \tag{5}$$

The exact density identity and the final density are

$$e^{-t_u}\mathrm{d}\operatorname{vol}_{h^+} = e^{-w}\mathrm{d}\operatorname{vol}_h, \qquad e^{-t^+}\mathrm{d}\operatorname{vol}_{h^+} = e^{u-w}\mathrm{d}\operatorname{vol}_h\,. \tag{6}$$

Confusing these two densities would change the physical operator. If h is complete, $h^+ \geq h$ is complete: a Cauchy sequence for h^+ converges in h, and local equivalence of smooth metrics gives convergence in h^+.

Proposition 2.1 (The source graph construction for fixed $c > 0$)**.** *On a complete smooth boundaryless (M, h), for every smooth w and each fixed $c > 0$, the source's global prescribed-graph argument produces a smooth solution of (4). Its local compactness constants may depend on c, h, w and the compact region. No uniform positive lower bound for l on all of M is asserted.*

Proof. We check the dependence on c in the construction, using the classical local elliptic estimates specified in the source. On a closed manifold replace both occurrences of w by τw, $0 \leq \tau \leq 1$. The solution at $\tau = 0$ is $u = 0$, and comparison with constants gives $|u| \leq \max|w|$. On the graph in $(M, h) \times \mathbb{R}$, use the normal $\nu = (l\nabla_h u, -l)$ and $Z = -\partial_z$. The prescribed weighted mean curvature is $p(x, z) = c(z - \tau w(x))$.

The normal-variation formula for the angle $l = \langle Z, \nu\rangle$ is

$$\mathcal{L}l = -c, \qquad \mathcal{L} = \Delta_\Gamma + \langle -\nabla(\tau w)^\top, \nabla_\Gamma\rangle + |\mathrm{II}|^2 + \mathrm{Ric}_{\mathrm{amb}}(\nu,\nu) + \mathrm{Hess}(\tau w)(\nu,\nu) + \partial_\nu p. \tag{7}$$

Here $\partial_\nu p = c(\nu_z - \tau \mathrm{d}w(\nu_h))$. This sign follows by translating in Z: the residual changes by $+p_z$, whereas its normal linearization is $-\mathcal{L}$. After discarding the nonnegative $|\mathrm{II}|^2$, the zeroth coefficient is bounded below by $-C$, uniformly along this compact continuation. At a minimum of l, $-c = \mathcal{L}l \geq -Cl$, hence $l \geq c/C > 0$ after increasing C. The flux derivative has eigenvalues l transversely to ∇u and l^3 along it. The equation is therefore uniformly elliptic on the attained compact range. Its linearization in u has zeroth term $-c$. The maximum principle and Schauder theory give invertibility and openness; the height, angle and higher interior estimates give closedness. Thus the closed-manifold solution exists for any fixed positive c.

For noncompact M, extend successively larger smooth compact domains to closed manifolds by modifying the data only outside the domain, making product collars and doubling. Orientability is not needed. The closed solutions require bounds independent of the extension on each fixed coordinate ball. For height, the source uses barriers $\pm(C + b)$, where b is smooth in the ball and comparable to $(R-r)^{-1}$ near its boundary. The mean-curvature flux of b and its weighted correction are bounded on the collar, whereas $c(C+b-w)$ is eventually larger with the required sign. Increasing C on the inner compact set completes the barriers. Comparison uses only the strict positivity of c.

For the angle estimate take a fixed nonnegative compactly supported cutoff ζ, and set $A = \max \zeta/l$ over its support. At a contact point of $l - \zeta/A$, the local version of (7), discarding $|\mathrm{II}|^2$, gives

$-c \geq -C'/A$. The coefficients and the first two graph derivatives of the base cutoff are bounded using the local height and prescribed-curvature bounds; all constants depend only on the fixed ball and data. Thus $A \leq C'/c$, giving a positive angle bound where $\zeta = 1$. Differentiation, interior Hölder estimates for first derivatives and Schauder bootstrapping give all higher local bounds. A diagonal subsequence solves (4) smoothly on M. No step singles out $c = 10$. This is the source's exhaustion and elliptic argument with the parameter dependence retained, not a replacement by a finite-dimensional inverse or a numerical existence assertion. □

3 A coupled square completion

We first retain the source's actual graph scalar identity. On the open set $s > 0$ write

$$\theta = 1 - l^2, \quad k = \sqrt{\theta} = ls, \quad e = \nabla_h u/s, \quad H = \operatorname{tr} J,$$
$$J = \text{second fundamental form of a level of } u.$$

and set

$$S = H - e(w), \qquad K = \frac{l^2}{\theta} e(\beta) = -e(\log k), \qquad M_1 = S - K. \tag{8}$$

Level norms use the metric induced by h, not a coordinate coefficient norm. The identities used below are

$$p := \operatorname{div}_{h,w}(ke) = kM_1, \tag{9}$$
$$\mathcal{D}(h^+, t_u) - \mathcal{D}(h, w) \geq \theta \left(|J|^2 + M_1^2 + K^2/l^2 - 2M_1 K + 2e(M_1) \right), \tag{10}$$
$$\Delta_{h^+} u - \langle \mathrm{d}t_u, \mathrm{d}u \rangle_{h^+} = lp + k\theta K/l, \qquad |\mathrm{d}u|^2_{h^+} = \theta. \tag{11}$$

For clarity, (10) is not an assumption of positivity of a finite curvature matrix. Here is its geometric derivation. In level coordinates $q = u$, $h = v_1^2 \mathrm{d}q^2 + \gamma_q$, $v_1 = 1/s$, Gauss and normal variation give

$$\begin{aligned} \mathcal{D}(h, w) = & -|J|^2 - S^2 - 2e(S) + \operatorname{Scal}_{\gamma_q} + 2\Delta_{\gamma_q} w - |\mathrm{d}_\gamma w|^2 \\ & - 2\Delta_{\gamma_q} v_1/v_1 + 2\langle \mathrm{d}_\gamma \log v_1, \mathrm{d}_\gamma w \rangle. \end{aligned}$$

Under $v_1 \mapsto v_1/l$, $w \mapsto w - \beta$, the tangential change is $f(2-f)|\mathrm{d}_\gamma \log s|^2 \geq 0$, where $f = -s\beta_s = \theta \in [0, 1]$. The new normal, second form and S are $le, lJ, lM_1 + K/l$. Substitution cancels the derivatives of K and gives (10). Formula (11) follows from (6) and $\nabla_{h^+} u = lke$.

Theorem 3.1 (Improved exact reserve)**.** *For $n \geq 1$, let u solve (4) with $c = 2$. Then*

$$\mathcal{D}(h^+, t^+) \geq \mathcal{D}(h, w) + (t^+)^2 - C_*(n), \qquad C_*(n) = 4n + \frac{4}{3}. \tag{12}$$

The output is the actual graph and drift (5). In particular, no change of physical norm and no positive lower bound on l is needed.

Proof. More generally, when $p = c(u - w)$, $e(M_1) = M_1 K + c(s - e(w))/k$ and $e(w) = H - M_1 - K$. Subtracting u from t_u in (10) and using (11) gives

$$\begin{aligned} \mathcal{D}(h^+, t^+) - \mathcal{D}(h, w) \geq & \, k^2 |J|^2 - 2ckH + \frac{k^2 K^2}{l^2} + 2k(c - \theta/l)K \\ & + p^2 + 2(c - l)p + 2c\theta/l - \theta. \end{aligned} \tag{13}$$

The first two completions, without dividing by $n-1$, are

$$k^2|J|^2 - 2ckH = |kJ - cI_{n-1}|^2 - (n-1)c^2, \tag{14}$$

$$k^2K^2/l^2 + 2k(c-\theta/l)K = (kK/l + cl - \theta)^2 - (cl-\theta)^2. \tag{15}$$

For $n=1$ the zero-dimensional level tensor and both terms of (14) are zero. Now $c=2$ and $t^+ = -p/2 - \beta$. Complete the pressure square *after* subtracting this exact drift square:

$$p^2 + 2(2-l)p - (p/2+\beta)^2 = \frac{3}{4}\left[p + \frac{4}{3}(2 - l - \beta/2)\right]^2 - \beta^2 - \frac{4}{3}(2-l-\beta/2)^2. \tag{16}$$

Equations (13)–(16) imply a loss of at most

$$C_n(l) = 4(n-1) + (2l-\theta)^2 + \theta + \beta^2 + \frac{4}{3}(2-l-\beta/2)^2 - 4\theta/l. \tag{17}$$

It remains to bound this scalar function on the whole interval $(0,1]$, including arbitrarily steep graphs. Write $C_n(l) = 4(n-1) + F(l)$. Expansion gives

$$F(l) = l^4 + 4l^3 + \tfrac{7}{3}l^2 + \tfrac{4}{3}l\log l - \tfrac{16}{3}l + \tfrac{4}{3}(\log l)^2 - \tfrac{8}{3}\log l + \tfrac{22}{3} - 4/l, \tag{18}$$

$$F'(l) = 4l^3 + 12l^2 + \tfrac{14}{3}l + (\tfrac{4}{3} + \tfrac{8}{3l})\log l - 4 - \tfrac{8}{3l} + 4/l^2. \tag{19}$$

The elementary inequality $\log l \geq 1 - 1/l$ yields

$$F'(l) \geq 4l^3 + 12l^2 + \tfrac{14}{3}l - \tfrac{8}{3} + \frac{4(1-l)}{3l^2} > 0. \tag{20}$$

For $l \leq 1/2$ the last term is at least $8/3$; for $l \geq 1/2$ the term $12l^2$ is at least 3. Thus $F(l) \leq F(1) = 16/3$ and $C_n(l) \leq 4n + 4/3$.

All calculations so far are on $s > 0$. The original output expressions are smooth across $s = 0$, so the inequality extends to its boundary by continuity. On the interior of $s = 0$, u is locally constant and the graph equation forces $u = w$ there, whence $t^+ = 0$. This proves the remaining case without dividing by k at a critical point. □

The simpler separate-square argument with $c = 2$ gives $4n + 11$: it keeps $p^2/2 + 4ks$, bounds $(t^+)^2 \leq p^2/2 + 2s$, and spends at most 2 when $s < 1$. Theorem 3.1 avoids that extra decoupling. Neither constant is claimed optimal among graph constructions, pressure choices, or attainable level geometries. Maximizing the completed scalar envelope does not prove that its algebraic equality cases are globally realizable.

4 An off-equation certificate without a global angle bound

Define, for an arbitrary smooth graph,

$$r = \operatorname{div}_{h,w}\left(\frac{\nabla_h u}{\sqrt{1 + |\mathrm{d}u|_h^2}}\right) - 2(u - w). \tag{21}$$

The following bounds concern this actual nonlinear residual and its physical gradient, not a finite list of sampled weak moments.

Theorem 4.1 (Uniform residual budget)**.** *Let h, w, u be smooth as above. If, everywhere on M,*

$$|r| \le \varepsilon \le 1, \qquad |\mathrm{d}r|_h \le \varepsilon_1, \tag{22}$$

then

$$\mathcal{D}(h^+, t^+) \ge \mathcal{D}(h, w) + (t^+)^2 - \left(4n + \frac{4}{3} + \frac{2}{3}\varepsilon + \frac{1}{3}\varepsilon^2 + 2\varepsilon_1\right). \tag{23}$$

Completeness of h makes h^+ complete, but the inequality itself is pointwise. No graph existence or global width conclusion follows merely from a finite verification of (22) on a bounded region.

Proof. Now $p = 2(u - w) + r$ and $t^+ = -(p - r)/2 - \beta$. In the derivation of (13), $e(M_1)$ gains $e(r)/k$, so the right side gains $2ke(r)$. Completing the p square with its modified drift gives the exact additional cost

$$C_n(l, r) = C_n(l) + a(l)r + r^2/3, \qquad a(l) = \frac{2}{3}(2 - l - 2\log l) > 0. \tag{24}$$

Indeed the pressure square now has center $\frac{4}{3}(2 - l - \beta/2 + r/4)$; expansion of its constant together with $(\beta - r/2)^2$ gives (24). Thus a more informative signed bound, before taking absolute values, is

$$\mathcal{D}(h^+, t^+) - \mathcal{D}(h, w) - (t^+)^2 \ge -C_n(l) - a(l)r - r^2/3 + 2\langle ke, \nabla_h r\rangle. \tag{25}$$

The vector $ke = l\nabla_h u$ is smooth even at $s = 0$.

We prove that $F(l) + \varepsilon a(l)$ increases on $(0, 1]$ for $0 \le \varepsilon \le 1$. Since $a'(l) = -\frac{2}{3}(1 + 2/l) < 0$, it is enough to show $F'(l) - \frac{2}{3}(1 + 2/l) > 0$. The same logarithm bound as before gives

$$3l^2\left[F'(l) - \tfrac{2}{3}(1 + 2/l)\right] \ge P(l) := 12l^5 + 36l^4 + 14l^3 - 10l^2 - 8l + 4 > 0. \tag{26}$$

Here positivity is a finite exact polynomial fact, not a sampled plot. For $x \in [0, 1]$, the degree-five Bernstein coefficients of $P(x/2)$ and $P((1 + x)/2)$ respectively are

$$\begin{aligned} &(4, 16/5, 43/20, 41/40, 9/20, 15/8),\\ &(15/8, 33/10, 269/40, 68/5, 131/5, 48). \end{aligned} \tag{27}$$

Every coefficient is positive, and the Bernstein basis is a nonnegative partition of unity. Expanding it gives (26) exactly. It follows that

$$C_n(l) + a(l)r + r^2/3 \le C_n(l) + \varepsilon a(l) + \varepsilon^2/3 \le C_*(n) + 2\varepsilon/3 + \varepsilon^2/3.$$

Finally $2\langle ke, \nabla r\rangle \ge -2\varepsilon_1$, since $|ke| = k \le 1$. Continuity covers boundary critical points. On an open set where u is constant, $p = 0$, $r = 2(w - u)$, $h^+ = h$ and $\mathcal{D}(h^+, t^+) = \mathcal{D}(h, w)$. There $(t^+)^2 = r^2/4 \le 1/4$, smaller than the claimed loss for $n \ge 1$. This covers the case not accessible by level coordinates. □

For larger or nonuniform residuals, (25) remains available. For example, if $l \ge l_0 > 0$ and $|r| \le E$ with no restriction $E \le 1$, the loss $C_* + a(l_0)E + E^2/3 + 2\varepsilon_1$ suffices. This latter certificate requires a genuine whole-domain angle bound; the main theorem does not. Neither statement estimates the distance to an exact solution. It certifies the reserve of the candidate graph itself.

5 Actual graph examples and a necessary derivative budget

Example 5.1 (A complete affine graph, with constant residual)**.** Take $M = \mathbb{R}^n$ with its Euclidean metric, $a \in \mathbb{R}$, $l = (1 + a^2)^{-1/2}$ and

$$u(x) = ax_1, \qquad w(x) = ax_1 + la^2/2 + r_0/2.$$

Then $\operatorname{div}_{h,w}(l\nabla u) = -la^2$ and the residual is exactly r_0. The output metric is constant, with first diagonal entry $1 + a^2$; $t^+ = la^2/2 + r_0/2 - \log l$ is constant. Hence

$$\mathcal{D}(h, w) = -a^2, \qquad \mathcal{D}(h^+, t^+) = 0.$$

This is an actual complete physical graph, not freely assigned values of the level variables. When $r_0 = 0$ it solves the $c = 2$ equation. For $a = 3/4$, $l = 4/5$ and $r_0 = 1/10$, its drift is $11/40 + \log(5/4)$, and the added certified residual cost is exactly $7/100$. This example has no positive initial spectral reserve and is not an instance proving the source's width hypothesis. It tests the normalization and the residual formula.

Example 5.2 (Small residual amplitude is insufficient)**.** Keep $a = 3/4$, $l = 4/5$, $\lambda = la = 3/5$, and fix $0 < \varepsilon \le 1$. For integers $N \ge 1$ on Euclidean $\mathbb{R}^n$ define

$$A_N = \frac{2\varepsilon}{4 + \lambda^2 N^2}, \qquad B_N = \frac{\lambda N \varepsilon}{4 + \lambda^2 N^2}, \qquad u = ax_1, \qquad w_N = ax_1 + \frac{la^2}{2} + A_N \sin(Nx_1) + B_N \cos(Nx_1).$$

Direct differentiation gives $2A_N + \lambda N B_N = \varepsilon$ and $2B_N - \lambda N A_N = 0$, so $r_N = \varepsilon \sin(Nx_1)$ and $\|r_N\|_\infty = \varepsilon$, although $\|\mathrm{d}r_N\|_\infty = \varepsilon N$. The graph angle is the fixed number $4/5$. At $x_1 = \pi/N$, $w_N' = a - NA_N$ and $w_N'' = N^2 B_N$. Since the output metric is constant and $t^+ = w_N - u - \log l$,

$$\begin{aligned} \mathcal{D}(h^+, t^+) - \mathcal{D}(h, w_N) &= -2(1 - l^2)N^2 B_N + (a - NA_N)^2 - l^2 (NA_N)^2 \\ &= -\frac{2(1 - l^2)\varepsilon}{\lambda} N + O(1). \end{aligned} \tag{28}$$

The output drift stays bounded at those points. Thus no reserve loss depending only on n, ε can replace the derivative budget in Theorem 4.1, even with a fixed positive angle. This is a counterexample to that proposed off-equation extension, not to the exact source graph theorem or the spectral-width statement.

For $n = 4$, the source loss is 603, the separate-square $c = 2$ loss is 27, and the coupled loss is $52/3$. If $\varepsilon = 1/10$ and $\varepsilon_1 = 1/100$, the robust loss is $5227/300$. These are verified costs in an inequality, not measured running times, sharp geometric obstructions, or computed uniform Urysohn widths.

6 Physical forms and what a finite Ram certificate can establish

For any smooth metric h and real smooth t, integration by parts on compactly supported ϕ gives the exact ground-state identity

$$\int (4|\mathrm{d}\phi|_h^2 + \operatorname{Scal}_h \phi^2)\, \mathrm{d}\operatorname{vol}_h = \int 4|\mathrm{d}\phi + \tfrac{1}{2}\phi\, \mathrm{d}t|_h^2\, \mathrm{d}\operatorname{vol}_h + \int \mathcal{D}(h, t)\phi^2\, \mathrm{d}\operatorname{vol}_h\,. \tag{29}$$

Equivalently, for $\phi = e^{-t/2}\psi$,

$$Q_h(e^{-t/2}\psi) = \int (4|\mathrm{d}\psi|_h^2 + \mathcal{D}(h, t)\psi^2) e^{-t}\, \mathrm{d}\operatorname{vol}_h\,.$$

The cross term in the first display integrates to $-2\int \phi^2 \Delta_h t$, which verifies both signs. Compact support avoids an unproved flux condition at infinity. On a domain with boundary the corresponding boundary term must instead be supplied, or the trace must vanish. These classical identities express the physical quadratic form; they are not newly claimed spectral theorems.

For example, if a candidate satisfies the whole-domain hypotheses of Theorem 4.1 and $\mathcal{D}(h, w) \geq L$, then

$$Q_{h^+}(\phi) \geq \int (L + (t^+)^2 - C_{\mathrm{rob}})\phi^2 \,\mathrm{d}\,\mathrm{vol}_{h^+}, \quad C_{\mathrm{rob}} = C_* + 2\varepsilon/3 + \varepsilon^2/3 + 2\varepsilon_1.$$

There is no need to invert a finite Gram matrix to obtain this conclusion. Conversely, a Gram matrix evaluated on a finite test family proves the form bound only on that family's span. It cannot supply the hypothesis for every compactly supported test.

The original local Ram material [6, 7] makes the same distinction in its dual testing and energy arguments: a finite set of weak residual pairings yields a restricted dual norm, ordinarily a *lower* bound on the full residual norm, unless the missing directions are controlled. In a coordinate chart the weak graph residual obeys

$$\int r\,\psi\, e^{-w} \,\mathrm{d}\,\mathrm{vol}_h = -\int \left\langle \frac{\nabla u}{\sqrt{1+|\mathrm{d}u|^2}}, \nabla\psi \right\rangle_h e^{-w} \,\mathrm{d}\,\mathrm{vol}_h - 2\int (u-w)\psi e^{-w} \,\mathrm{d}\,\mathrm{vol}_h\,. \tag{30}$$

Recovering r and especially $\mathrm{d}r$ from finitely many such moments requires a genuine approximation class, all omitted terms, correct coordinate densities, and an independent remainder bound. A small matrix residual is not such a bound.

This is particularly important because r contains second derivatives of u, and $\mathrm{d}r$ generally contains third derivatives. A rigorous panel certificate for Theorem 4.1 would need control of these derivatives, the metric and density coefficients, patching across charts, and the whole noncompact complement. The original TR8 quasilinear differentiation keeps the coefficient derivatives; omitting them changes the operator. Its bounded-inverse and Newton statements also require an actual operator between stated infinite-dimensional spaces with the correct trace or gauge. None is inferred here from a truncated stiffness matrix.

7 Signed pressure and the unchanged global interface

7.1 Why the sign of the drift matters

The new drift t^+ need not be nonnegative. The source's later pressure $F(x,t)$ is odd in the discrepancy x and defined for all $t \in \mathbb{R}$, with poles at $x = \pm S_0$. With $N(t) = \sqrt{t^2+T^2}$ its required bounds include

$$\begin{gathered} F_x \geq A_0 N, \quad |F_t| \leq \epsilon|F|, \quad FF_t \leq \epsilon F_x, \\ F_x \leq b_0 N^2 \quad \text{or} \quad -FF_t \geq P_0 F_x, \qquad F_x \leq b_0 N^2 + \epsilon F^2. \end{gathered} \tag{31}$$

This is a different, nonlinear pressure from the preprocessing $2(u-w)$. We do not replace it by a linear law. In particular $F = N(t)\tan(kx)$ alone fails the alternative: for $t > 0$ near the positive pole, F_x/N^2 is unbounded and $FF_t > 0$.

The source constructs the inverse $x = X(f,t)$, $f > 0$, using $q = f^{2/3}$, $y = t/q$, $z = T/q$ and a smooth positive rounded maximum $G(y,z)$ equal to $\sqrt{y^2+z^2}$ outside a compact set. Precisely,

$$X = \frac{1}{k}\arctan\frac{\sqrt{q}}{G}, \quad A_G = G + 2(yG_y + zG_z), \quad X_f = \frac{A_G}{3kq^2(1+G^2/q)}, \quad fX_t = -\frac{G_y}{k(1+G^2/q)}. \tag{32}$$

Positivity of A_G permits inversion. Where $G_y < 0$, the identity $FF_t/F_x = -fX_t$ supplies the favorable negative mixed sign. Outside that region the reserve controls F_x instead. The auxiliary function uses an average of $\max\{R, \lambda m(y)\}$ over a smooth density in $1 < \lambda < 2$, with $m(y) = D_G e^{-\arctan(y)/8}$. This preserves the needed smoothness and derivative inequalities. The constants defining G are fixed before T; only afterward is T made large. Odd extension in x preserves all products in (31); no restriction $t \geq 0$ is legitimate. These mechanisms are retained, not new contributions.

7.2 Trace losses, local finiteness, and initialization

The source's local stretching lemma, retained here as an input, gives $h' \geq h$, $t' \geq t$, compact support of the change and

$$\mathcal{D}(h', t') \geq \mathcal{D}(h, t) + (V(t') - V(t))_+ - \big[\delta(\operatorname{tr}_h h_0 - \operatorname{tr}_{h'} h_0) + d\big]V(t'), \quad V(t) = t^2 + T^2. \tag{33}$$

It requires the specified separation in the complete background h_0 and supplies a factor $L_* = 1.004$ increase. Although $t' \geq t$, $V(t') \geq V(t)$ need not hold. The source correctly uses the positive part and the elementary inequality $a_+ \geq \Lambda a$ for $0 \leq \Lambda \leq 1$. With

$$\Lambda = 1 - \delta(n - \operatorname{tr}_H h_0) - \sum d_i, \qquad \sum d_i < \delta,$$

the invariant $\mathcal{D}(H, z) \geq b + \Lambda V(z)$ survives a whole round, even without a uniform overlap bound. This is why deleting the positive part on the basis of monotonicity of t would be erroneous.

Only the initial reserve cost changes in the present report. Keeping the source's $W, T, D_{\mathrm{nat}}(n)$ and δ unchanged, choose

$$R_0 \geq 1, \qquad R_0^2/2 \geq C_*(n) + T^2, \qquad W R_0 \geq D_{\mathrm{nat}}(n). \tag{34}$$

For an approximate graph certified in the working metric g/R_0^2, replace C_* by C_{rob}, with both residual norms measured in that working metric. Applying preprocessing to g/R_0^2 gives

$$\mathcal{D}(h^{[0]}/R_0^2, t^{[0]}) \geq R_0^2 + (t^{[0]})^2 - C_* \geq R_0^2/2 + V(t^{[0]}).$$

The exact solution is available through Proposition 2.1; the approximate version is only a conditional alternative. Subsequent rounds take $R_j = L_*^j R_0$ and retain the unchanged condition $2(n+1)\delta < 1 - L_*^{-2}$. No other small parameter is relaxed by the improved preprocessing constant.

The source fixes compact original supports for every center label; these form a locally finite family. Each label eventually exceeds its prescribed separation target, hence is used only finitely often. Consequently every compact region stabilizes after finitely many changes, yielding smooth complete limiting data with $\mathcal{D}(h, t) \geq 1/2$. This is stronger than mere pointwise convergence of metrics and is an essential part of the imported assembly.

Finally, a compactly supported partition of unity decomposes $-t$ into pieces τ_ν. Splitting each into m_ν equal log warps preserves $P = \sum_\alpha p_\alpha = -t$ and permits $\sum_\alpha |\mathrm{d}p_\alpha|_h^2 \leq 1/10$. Thus

$$R(h, (p_\alpha)) = \mathrm{Scal}_h - 2\Delta_h P - |\mathrm{d}P|_h^2 - \sum_\alpha |\mathrm{d}p_\alpha|_h^2 \geq 2/5. \tag{35}$$

The splitting identity is classical and appears in Gromov's stabilization discussion [3]. Repeating each support only finitely many times preserves local finiteness; an uncontrolled infinite product of warps is not being substituted.

7.3 What still gives width

The separate source partition interface assumes the complete enlarged metric, the locally finite log warps, (35), and all distances

$$\operatorname{dist}_h(A_i, B_i) > 10 + 2\pi\sqrt{2/p_i^*}, \qquad p_i^* > 0, \qquad \sum_{i\geq 0} p_i^* = 1/20.$$

For a maximal D-separated set in the original metric, $D = WR_0$, it then gives whole fibers of original diameter at most $4D + 2$. The relevant partition and topological proof from the companion manuscript has not been independently re-established here. Consequently the expression $4WR_0 + 2$ with (34) is a conditional replacement in the source's constant chain, not a numerical best width, a new topological theorem, or a certificate of that companion. Large T or D_{nat} may dominate and erase much of the practical effect of the reserve improvement.

8 Status, advantages, and remaining work

The assigned text is a publisher preprint in the captured repository, not a peer-review verdict. Fresh public primary checks on October 9, 2026 included Gromov's 2024 stabilization paper [3], with its relevant scalar formulas and spectral discussion read, and the metadata/abstracts of Liokumovich–Wang [4] and Kumar–Sen [5]. The former concerns complete three-manifolds and a singular foliation; that statement is not silently substituted for whole fibers in dimension $n \geq 4$. The latter's current v2, February 1, 2026, reports counterexamples to the *macroscopic* scalar-curvature width conjecture in dimensions at least four and noncontinuity under collapsing. Its hypothesis is not the pointwise or physical spectral bound (2), so no contradiction with the assigned claim is inferred. External full proofs and an exhaustive priority search were not undertaken.

The earned increment is the coupled source-specific completion and its uniform residual extension, not the maximum principle, Bernstein positivity, ground-state transformation, or a new Ram spectral theorem. It has three useful features: the actual induced metric and density are retained; the constant is uniform even when the graph slope is unbounded; and the signed residual identity exposes exactly which defect derivatives matter. The conditional effect on initialization is explicit.

The costs are equally concrete. The graph equation still has to be solved globally, or its residual bounds proved globally. A bound on dr costs one more derivative than a bound on r. The global pressure construction, transition cutoffs, local stretching, local finiteness and partition theorem are untouched. Smaller reserve loss need not produce a comparably smaller final width constant. The envelope is not an optimality theorem, and no numerical speedup has been measured.

There are several specific extensions worth investigating. One can complete the pressure square for other $c > 1$ and optimize the resulting one-variable envelope, but must retain the actual output drift and critical-point cases. A weak-form residual certificate might integrate the directional derivative in (25) by parts, but would need the correct final density in (6), boundary or infinity flux, and additional mixed terms; our C^1 result does not already prove such an extension. Finally, an effective global width bound would require quantitative control of all subsequent pressure and partition constants, not only the preprocessing improvement.

9 Read scope and reproducible checks

The owned literal-read ledger records PRE hashes before the first displays and matching POST hashes. Public coverage is all nine text files, 2139 lines: main 61, introduction 169, interface 119,

preprocessing 534, pressure 248, stretching 575, assembly 252, bibliography 163 and README 18. There are no unread lines in these assigned files. Rereads are not added to the unique-line total.

Fresh confidential original reads were [6], lines 139–205, 233–361, 380–442 and 637–745 (368 lines), and [7], lines 53–235, 270–372, 492–558 and 841–872 (385 lines). The selected private files contain 3296 lines in total, of which 2543 were not read; exact complements are in the ledger. TR8's four included table files and other referenced private works were not proof-read. No whole-corpus or inherited-read claim is made.

The owned executable `M649-checks.py` passes 58 exact or symbolic cases with 167 assertions, and zero floating-point cases. It verifies the square identities, the exact Bernstein expansions and positivity, physical level-tensor completions including $n = 1$, actual affine-graph residuals, the oscillatory negative control, and reserve/initialization budgets. The finite checks supplement the written proofs; they do not verify noncompact PDE existence, the partition interface, or the width claim. Parent and foreign cases are excluded. The receipt records the executed script hash; compilation, page inspection and the final read-only freeze check have separate owned receipts.

References

[1] OpenAI, *Spectral scalar curvature and uniform Urysohn width*, manuscript dated October 5, 2026, captured repository source, main and six section files. Public locator: publisher manuscript directory. The assigned source is read locally in full; byte identity is not proof validation.

[2] OpenAI, *Positive scalar curvature and uniform codimension-two width*, companion cited by the assigned manuscript. Used only through its explicitly imported partition interface; full companion proof not independently audited in this report.

[3] M. Gromov, *Product Inequalities for $\mathbb{T}^{\times}$-Stabilized Scalar Curvature*, SIGMA 20 (2024), 038, 25 pp. Primary publication. Relevant initial scalar and spectral discussion read, not all proofs.

[4] Y. Liokumovich and Z. Wang, *On the waist and width inequality in complete 3-manifolds with positive scalar curvature*, arXiv:2308.04044, v1, August 8, 2023. Fresh primary abstract/version reading only.

[5] A. Kumar and B. Sen, *Urysohn width and macroscopic scalar curvature*, arXiv:2601.14669v2, February 1, 2026. Fresh primary abstract/version reading only.

[6] M. Subbarao, original confidential local `RamPatsCore50-TRs`, `TR-core-pat-4/TR-core-pat-4.tex`. Physical metric, dual restriction, panel and energy passages listed in Section 9; other portions not audited.

[7] M. Subbarao, original confidential local `RamNonlin-IE-PDE-astra`, `TR8-lin-non-lin/TR8-lin-non-lin.tex`. Domain, physical operator, quasilinear differentiation and residual passages listed in Section 9; included tables unread.